

LYNRED

Data challenge 2026

Defective columns

Applications Lab

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KEY FIGURES AND SHAREHOLDERS



GLOBAL IR INDUSTRY LEADER

offering the largest
Infrared product portfolio



RESEARCH PARTNER
CEA LETI- ONERA- III-V LAB



80%
EXPORT



> 133 PATENT FAMILIES
> 680 PATENTS FILED



15% REVENUE
INVESTED in R&D



> 2 MILLION DETECTORS
SHIPPED SINCE 1986

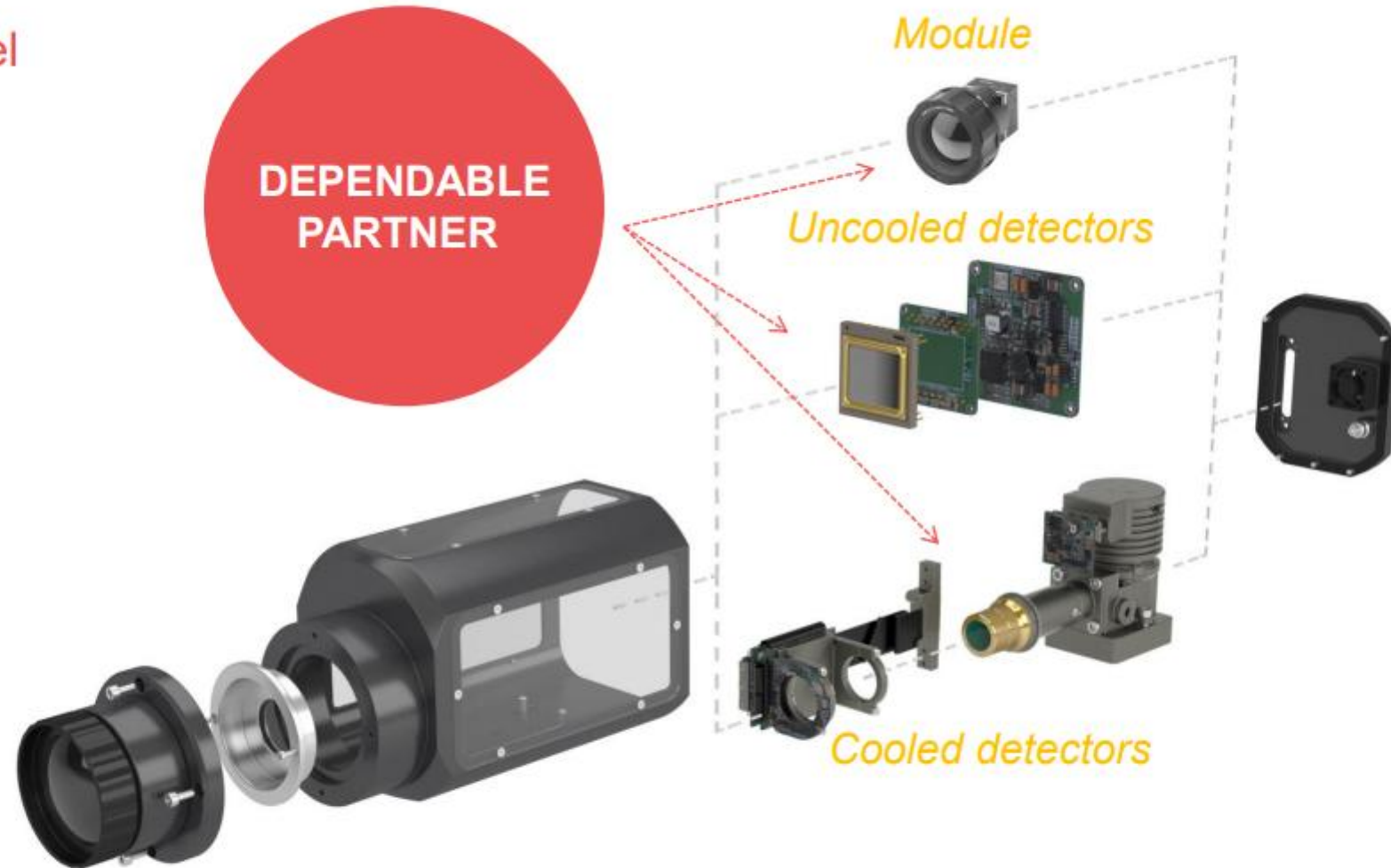


FULL INFRARED
SPECTRUM



Working on automotive
IATF 16949 certification

About LYNRED
Business Model



Target markets

Designed to deliver what your
market demands

AEROSPACE



DEFENSE



SECURITY,
SURVEILLANCE
& SAFETY



THERMOGRAPHY



LEISURE &
OUTDOOR



MACHINE
VISION



TRANSPORT &
MOBILITY



ENVIRONMENT



Context

Infrared image with high scene dynamic

- ▶ Good weather conditions
- ▶ High scene contrast
- ▶ → Few defects visible



High quality infrared image

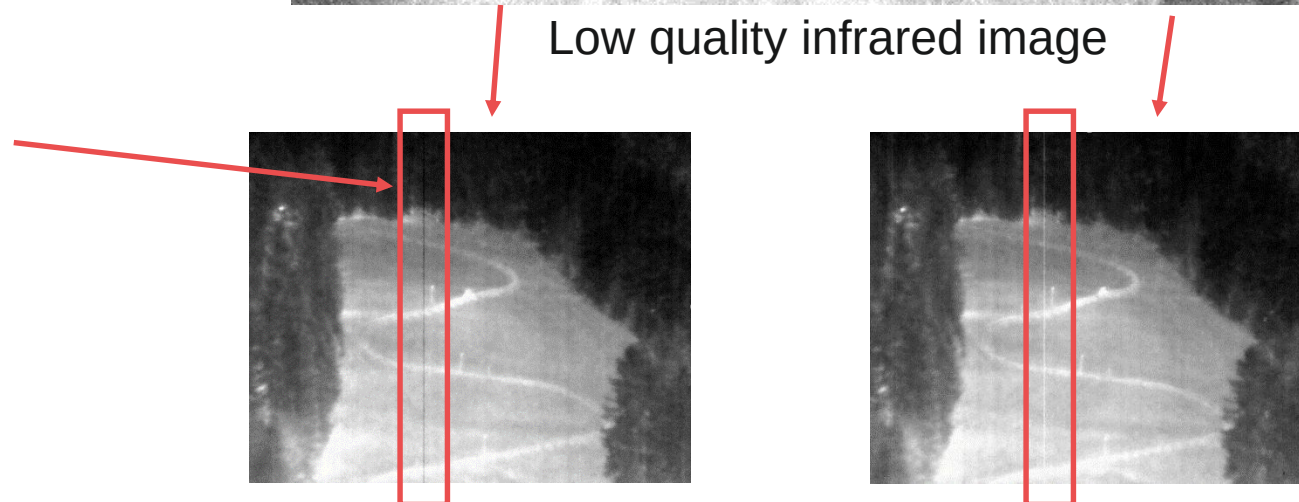
Context

Infrared image with low scene dynamic

- ▶ Bad weather conditions
- ▶ Low scene contrast
- ▶ → Defects visible
 - ❑ Temporal noise
 - ❑ Spatial noise
- ▶ Correlated defects more visible
 - ❑ Example : defective columns



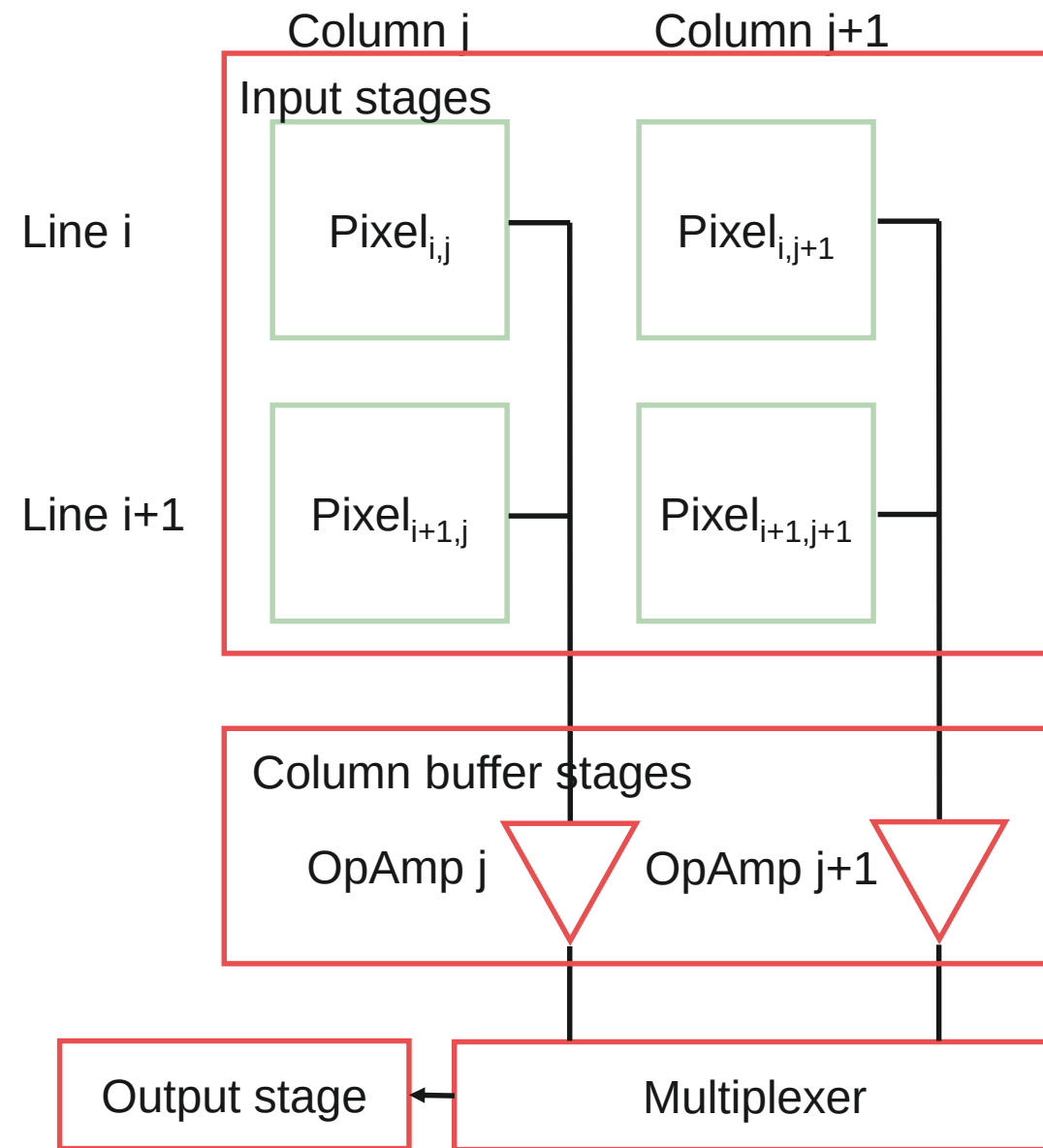
Low quality infrared image



Defective columns definition

Physical origine of the defect

- ▶ Some operational amplifier (OpAmp) more sensible
- ▶ One operational amplifier for one column or One operational amplifier for four columns
 - ❑ Data OpAmp1column/
OpAmp4columns



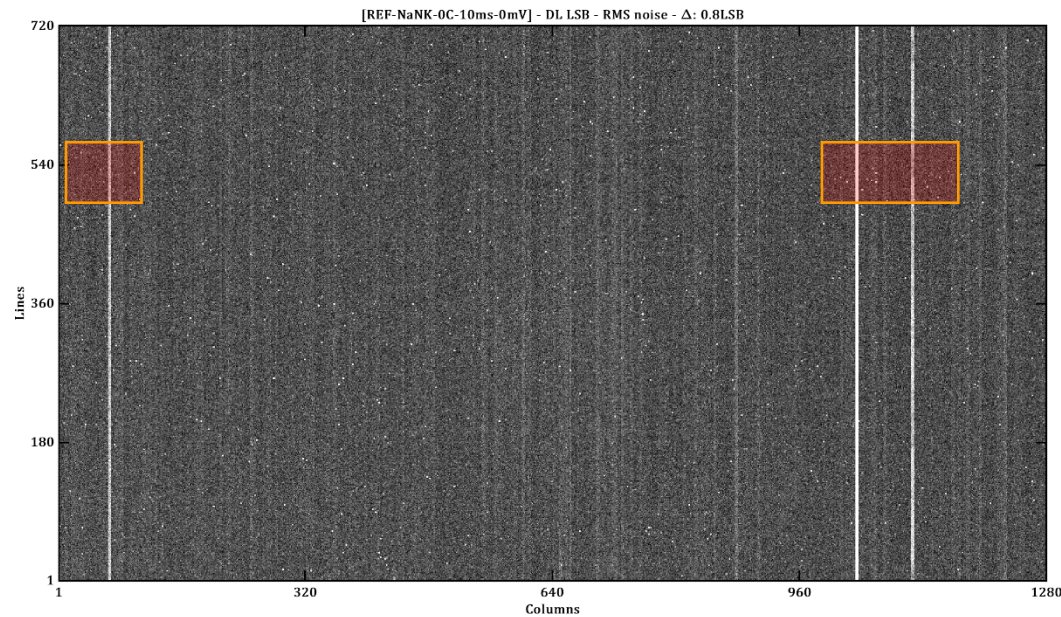
General architecture of readout circuit

Defective columns definition

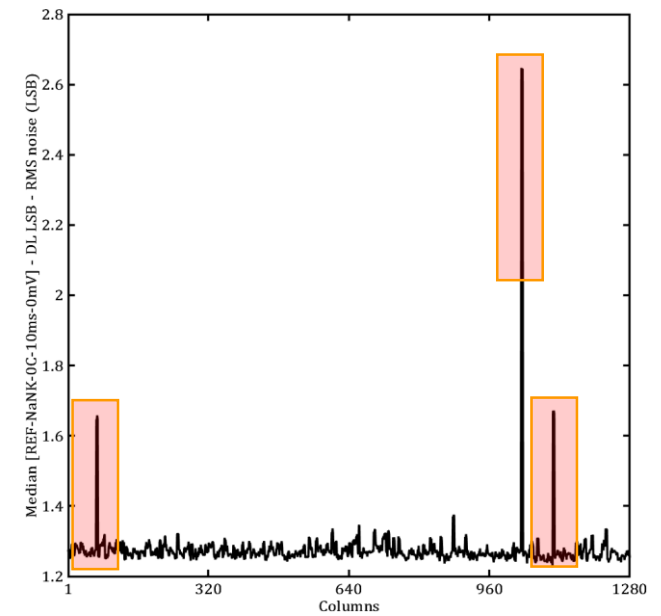
Defective columns type: noisy columns

- ▶ Noise of the column is almost always higher than the noise of the background

Noise Image : standard deviation along temporal direction



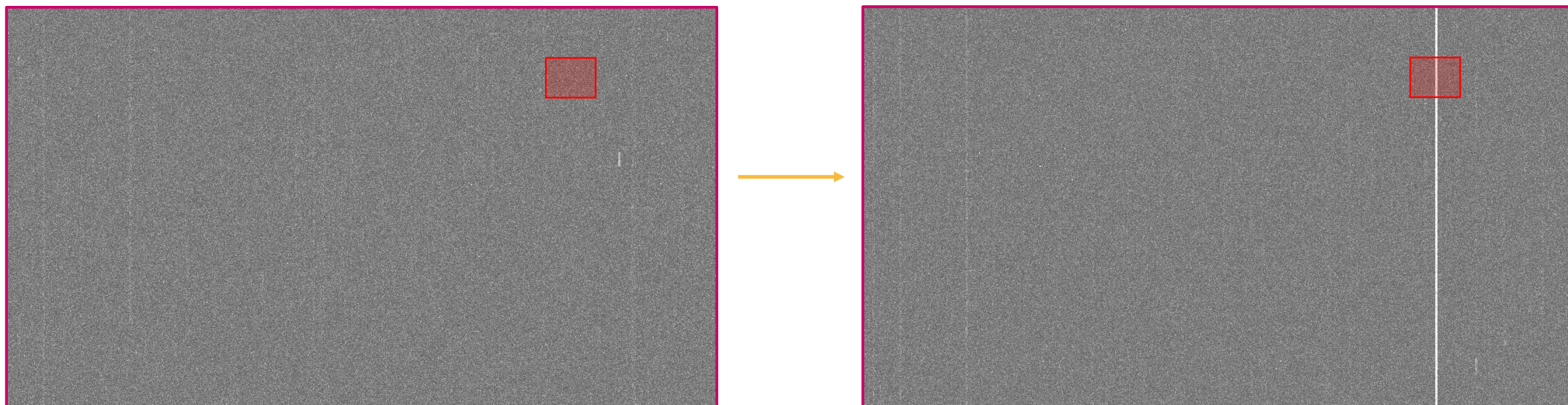
Median noise of each column



Defective columns definition

Defective columns type: blinking columns

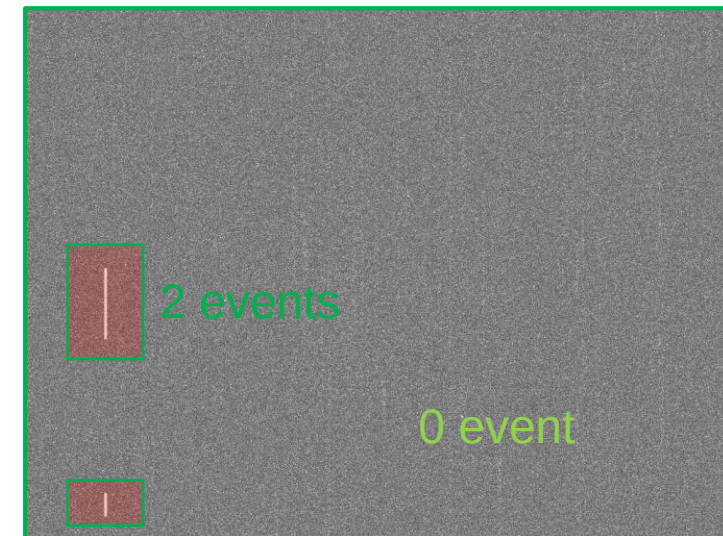
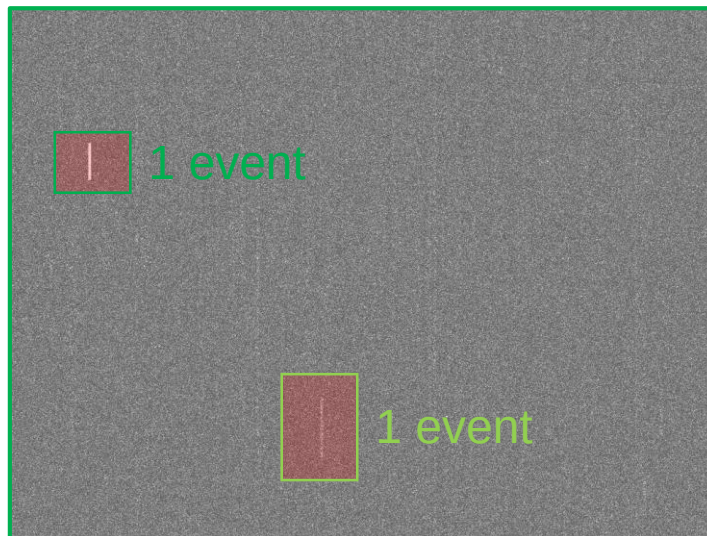
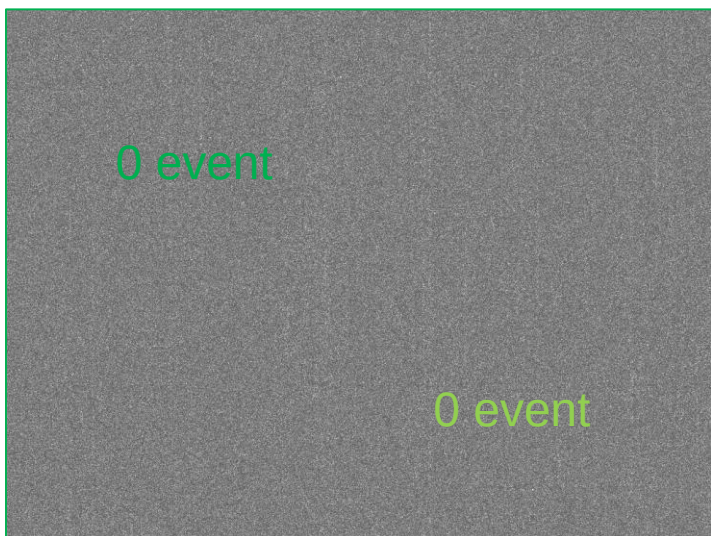
- ▶ Digital level change from time to time
 - ❑ Sometimes at high frequency, sometimes at low frequency
- ▶ Blinking column can be noisy and blinking



Defective columns definition

Defective columns type: fragmented blinking columns

- ▶ Digital level changes in time AND spatial directions
- ▶ Always fast changes : it changes at each frames
- ▶ One, two or more events per frame



Data challenge

Goals

- ▶ Detect defective columns
- ▶ Correct defective columns

Data challenge

Dataset

- ▶ Image format: 16 bits .png
- ▶ Train and test datasets
 - ❑ Simulated images without real defective columns
 - ❑ Simulated images without real defective columns and with simulated defective columns
 - Metadata in JSON file
 - Type of defective columns
 - Spatial and temporal position
 - Amplitude
 - ❑ Train dataset given
 - ❑ Test dataset not given → we calculate your score
- ▶ Create /result in each image folder to run metrix.py
 - ❑ Example: \train\VGA\sequence_1\high_dyn\result → your corrected images

Data challenge

Detection metrics

- ▶ A defective column detected = True Positive (TP)
- ▶ A non defective column detected = False Positive (FP)
- ▶ A defective column not detected = False Negative (FN)

- ▶ $Precision = \frac{TP}{TP+FP}$: how many of the defective columns predicted were correct
- ▶ $Recall = \frac{TP}{TP+FN}$: how many of the defective columns you managed to detect

- ▶ $F1score = 2 \times \frac{Precision \times Recall}{Precision+Recall}$: Combines Precision and Recall

Data challenge

Correction metrics

► Correction

$$\square RMSE_{defective} = \sqrt{\frac{\sum_{i=1}^N (\hat{y}_i - y_i)^2}{N}}$$

- With N the total number of defective pixels in all defective columns
- With y_i the ground truth value
- With $\hat{y}_i$ the corrected value estimated by the algorithm

$$\square RMSE_{ok} = \sqrt{\frac{\sum_{i=1}^M (\hat{y}_i - y_i)^2}{M}}$$

- With M the total number of non defective pixels on the image
- With y_i the ground truth value
- With $\hat{y}_i$ the corrected value estimated by the algorithm

Data challenge

Final metric

$$\begin{aligned} \text{FinalScore} = & 0.34 \times F1score \\ & + 0.33 \times \left(1 - \min \left(\frac{RMSE_{defective}}{40}, 1 \right) \right) \\ & + 0.33 \times \left(1 - \min \left(\frac{RMSE_{OK}}{40}, 1 \right) \right) \end{aligned}$$

- ▶ Higher is better with a maximum value of 1
- ▶ Detection performance, RMSE_defective and RMSE_ok account for ~33% of the score each
- ▶ RMSE's are capped at 40 digital levels meaning that a mean error of more than 40 on the pixel values leads to the worse possible RMSE score of 0.

Deliverables

- ▶ Code to run on \train to get \result with corrected images
 - ❑ Deadline 2026-06-02 20:00
- ▶ Report and presentation
 - ❑ Deadline 2026-06-04
 - ❑ Presentation of the solution
 - ❑ Metrics on \train with your comments
 - Are you happy with your correction?

Advices

- ▶ Develop first on small data: ie part of VGA files
- ▶ Always take a look on images
- ▶ Focus on quick V0 to get a base line
- ▶ Take all the support you need but do not use LLM

Merci
POUR VOTRE ATTENTION



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